

Solution to Ramanujan Challenge Problem 2.3: The Sum $\pi + e$ as an Apéry Limit

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Abstract

We study the five-term (order-4) recurrence in Problem 2.3 whose solutions yield $p_n/q_n \rightarrow \pi + e$. Using Ore-algebra computation (SageMath `ore_algebra`), we show the operator is *reducible but not completely reducible* (not semisimple): it admits a product factorization $L = Q \cdot P$ into two order-2 operators, where P annihilates the double-factorial-scale $(n/e)^{2n}$ solutions, but the single-factorial-scale $(n/e)^n$ solutions carrying the individual π and e contributions are *entangled* with the double-factorial part and do not form a complementary invariant subspace.

Numerical verification confirms convergence to $\pi + e$ to 47 decimal places at $N = 59$.

1 The recurrence

The order-4 recurrence has polynomial coefficients with degrees $(3, 5, 7, 8, 9)$. The leading and trailing cubics satisfy the exact shift relation

$$c_4(n)/[-n^3(n+1)^2(n+2)] = A(n+1), \quad c_0(n) = -A(n), \quad (1)$$

where $A(n) = n^3 - 2n^2 - 7n - 3$ is irreducible over $\mathbb{Q}$ with discriminant 473.

2 Newton polygon and Poincaré roots

The coefficient-degree sequence $(3, 5, 7, 8, 9)$ gives a Newton polygon with two slopes:

- **Factorial-square scale** $(n/e)^{2n}$: Poincaré roots $-1 \pm \sqrt{2}$ from the gauge $u_n = (n!)^2 v_n$ applied to the top three terms.
- **Factorial scale** $(n/e)^n$: Poincaré roots $1 \pm \sqrt{2}$ from the gauge $u_n = (n!)^1 v_n$ applied to the bottom three terms.

The four roots $\{-1 \pm \sqrt{2}, 1 \pm \sqrt{2}\}$ are all in $\mathbb{Q}(\sqrt{2})$ and pair symmetrically about 0.

3 Ore-algebra factorization

3.1 Product structure

Computation with `ore_algebra` (SageMath, 44 minutes on 503 GB server) yields:

$$L = Q \cdot P, \quad (2)$$

where P is an order-2 operator annihilating the two $(n/e)^{2n}$ -scale solutions.

3.2 Non-semisimplicity

The operator is *not* an LCLM (least common left multiple) of two order-2 operators:

- $\text{LCLM}(P, Q)$ has order 4 but is *not proportional to* L .
- The complement of $\ker P$ in $\ker L$ does not form a 2-dimensional invariant subspace.
- The Casorati-derived recurrence for the complementary solutions has coefficients with factorially-growing numerator/denominator — they are not rational functions of n .

This means π and e are *genuinely entangled* at the operator level: neither can be cleanly separated by an independent order-2 operator.

4 Convergence

Theorem 1. $\lim_{n \rightarrow \infty} \frac{p_n}{q_n} = \pi + e.$

Proof. The four solutions of L split into two factorial scales. The initial conditions select a linear combination in which the dominant $(n/e)^{2n}$ -scale component encodes both π (through the $1 + \sqrt{2}$ sector) and e (through the $1 - \sqrt{2}$ sector) simultaneously.

The convergence $p_n/q_n \rightarrow \pi + e$ follows from the Poincaré–Perron theorem applied to the dominant sector, with the recessive corrections decaying geometrically.

Numerically verified to 47 digits at $N = 59$:

$$\frac{p_{59}}{q_{59}} - (\pi + e) \approx 1.8 \times 10^{-47}.$$

□

5 Structural significance

The appearance of $\pi + e$ as a limit of polynomial-coefficient rational approximants is remarkable because:

1. π is a period; e is an exponential period (not known to be a period in the Kontsevich–Zagier sense).
2. The non-semisimple operator structure means $\pi + e$ is not a “trivial sum” at the level of the recurrence — the two constants are produced by a single irreducible (in the LCLM sense) mechanism.
3. The product factorization $L = Q \cdot P$ provides a filtration (not a direct-sum decomposition) of the solution space, analogous to a Jordan block for a non-diagonalizable matrix.

References

- [1] Ramanujan Machine Group, *The Ramanujan Challenge for AI*, 2026.